

BT IsAtLocation: use the stock UE5 decorator

IN SHORT

UE 5.8 already ships `UBTDecorator_IsAtLocation`. Use it directly on LCM Nav3D pawns. No LCM Nav3D-specific variant exists or is needed.

Why we didn't ship `LCM_BTDecorator_IsAtLocation`

A stock-reuse audit found that `UBTDecorator_IsAtLocation` already does what an LCM Nav3D-side variant would do:

- Reads a `Vector` or `Object` blackboard key
- Computes distance from the pawn to the key
- Returns true if within `AcceptableRadius`
- Optional `bPathFindingBasedTest` reuses the active path-following result

This is the canonical "did the AI reach the location" predicate. Re-shipping it on the LCM Nav3D side would be redundant and force buyers to pick between two identical nodes.

Three-step usage on an LCM Nav3D pawn

1. Drop the stock decorator on a Sequence / Selector node in your BT graph. Right-click → Add Decorator → "Is At Location".
2. Set the blackboard key. Vector key (a goal position) or Object key (an AActor whose location is the goal; `bUseNavAgentGoalLocation` projects to nav-agent location). For LCM Nav3D flying pawns where you want the literal world position, use Vector.
3. Set `AcceptableRadius` to your "we count this as arrived" distance in cm. Typical: 50-150 cm.

The decorator works on every LCM Nav3D pawn that has an AIController, with no LCM Nav3D-specific configuration needed.

When to use the path-finding variant

`UBTDecorator_IsAtLocation::bPathFindingBasedTest` makes the decorator consistent with whatever was used to reach the location, useful when the move was via `UBTTask_MoveTo` (Recast) so the "arrived" check matches.

For LCM Nav3D moves, this option will fall through to Recast even if your goal was inside an LCM Nav3D volume. Leave it OFF and use straight geometric distance. With the NavSystem bridge registered, the stock `bPathFindingBasedTest` routes through LCM Nav3D automatically when the goal is in an LCM Nav3D-managed region.

Related LCM Nav3D-unique decorators

For predicates that **don't** have a stock equivalent, see the LCM Nav3D-unique decorators we did ship:

- `LCM_BTDecorator_PathLengthBelow` : true-3D LCM Nav3D path-length check (stock `DoesPathExist` is binary + Recast)
- `LCM_BTDecorator_HasLineOfSight` : voxel-aware Lazy Theta* LOS (stock uses raycasts only)
- `LCM_BTDecorator_IsInCover` : composite SweepPenalty + LOS predicate
- `LCM_BTDecorator_ChunkLoaded` : SVO streaming probe
- `LCM_BTDecorator_ThreatVisibleViaPerception` : AIPerception + LCM Nav3D LOS composite

These are genuinely additive surfaces; "IsAtLocation" is not, so use stock.

Chapter 12, section 2 of the LCM Nav3D User Manual.

The complete manual is `LCM_Nav3D_Manual.pdf`, in this folder alongside every other document in the set. Start from `README.pdf`.

LCM Nav3D: True Volumetric 3D Navigation for Unreal Engine. Version 2.05 · Unreal Engine 5.8.

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